TELANGANA TOURISM DATA ANALYTICS

1. PROJECT OVERVIEW

This project analyses annual domestic and foreign tourist visitor data for Telangana from 2014 to 2023. It uses MySQL for data querying and analysis, Microsoft Excel and VBA for report automation, and Microsoft Power BI for interactive visualization.

The project aims to identify visitor trends, calculate year-over-year growth, compare domestic and foreign visitors, and present findings that support tourism-related planning.

2. PROBLEM STATEMENT

Tourism datasets contain information that can help identify changes in visitor numbers over time. Without structured analysis and visualization, it can be difficult to identify growth patterns, significant declines, and unusual values.

This project develops a data analytics workflow to organize tourism data, calculate annual growth, automate summary reporting, and visualize trends through a dashboard.

3. OBJECTIVES

- Analyze annual domestic and foreign visitor counts.
- Calculate total visitors for each year.
- Measure year-over-year percentage growth or decline.
- Identify the years with the highest and lowest visitor totals.
- Automate Excel reporting using VBA.
- Build an interactive Power BI dashboard.
- Present preliminary business insights and recommendations.

4. TOOLS AND TECHNOLOGIES

- **MySQL:** Database querying, aggregation, and year-over-year analysis.
- **Microsoft Excel:** Data presentation, calculations, and reporting.
- **VBA:** Automated formatting, summary generation, and chart creation.
- **Microsoft Power BI:** Interactive dashboard, KPI cards, charts, and year filtering.

5. METHODOLOGY

Data Preparation

The annual visitor dataset was organized by year, domestic visitors, and foreign visitors. Total visitors were calculated by adding domestic and foreign visitors.

SQL Analysis

MySQL was used to create an analytical view and calculate year-over-year growth using a window function.

Excel and VBA Automation

Excel was used to display the dataset and summary metrics. VBA macros automated report formatting, summary generation, and annual trend chart creation.

Dashboard Development

Power BI was used to create KPI cards, annual trend charts, visitor comparisons, a year slicer, and a growth table.

Business Insights

The resulting figures were examined to identify unusually high or low visitor counts and changes between years.

6. KEY FINDINGS

In the current project dataset:

- The highest annual visitor total occurs in 2016.
- The lowest annual visitor total occurs in 2017.
- The largest positive year-over-year change occurs in 2016.
- The largest negative year-over-year change occurs in 2017.
- Domestic visitors represent the majority of the recorded visitor totals.

7. LIMITATIONS

- The dataset used in this demonstration has been modified. Its figures should therefore be treated as project data rather than verified official Telangana tourism statistics.
- The extreme 2016 value substantially affects the calculated growth rate and visual trend. The data should be validated against an authoritative source before using the results for real-world planning or forecasting.
- The analysis identifies patterns in the recorded data but does not establish the causes of visitor changes.

8. CONCLUSION

The project demonstrates an end-to-end data analytics workflow using MySQL, Excel VBA, and Power BI. It combines structured data analysis, automated reporting, and interactive visualization to make annual tourism trends easier to examine.
The workflow can be extended with validated tourism statistics, additional geographic and seasonal data, and more detailed analysis to support evidence-based tourism planning.